

MIDTERM SUMMER 2022 SOLUTION
Question 0)

1) When $V_i = \text{logical high}$.

State of T_1 : saturation because we are

applying a logical high input at the base voltage.

State of T_3 : saturation same reason as

above

State of T_2 : cutoff Because base voltage

will be less than 0.2 V which is

not enough to drive the T_2 transistor

$$V_{BE} = \frac{0.0 - 0.0}{1000} = 0 \text{ V}$$

$$V_{CE} = 0.0 \text{ V}$$

(b) Initial = 0 state

$$I = \frac{V_0 - 0.8}{400 \Omega} \quad \text{for } N=6$$

at $t=0$ Initial = 0 state

$$I_0 = 6 \times \frac{V_0 - 0.8}{400 \Omega}$$

at $t=0$ Initial = 0 state

$$I_0 = (6+1) \times \frac{5 - (V_0 + 0.7)}{(640 + 400) \Omega}$$

$$41 \times \frac{4.3 - V_0}{1040 \Omega}$$

at $t=0$ Initial = 0 state

Now $I_0 = I_0$

$$41 \times \frac{4.3 - V_0}{1040} = 6 \times \frac{V_0 - 0.8}{400}$$

$$V_0 = 3.935 \text{ V}$$

(c)

Verification of T₂ operating mode:

We need to prove $V_{CE} > 0.2 \text{ V}$ for

T_L

$$\text{Now } I_B = \frac{5 - (V_B + V_{BE})}{1040} = \frac{5 - (2.335 + 0.7)}{1040}$$

$$= 0.928 \text{ mA}$$

$$\text{So } I_C = \beta I_B$$

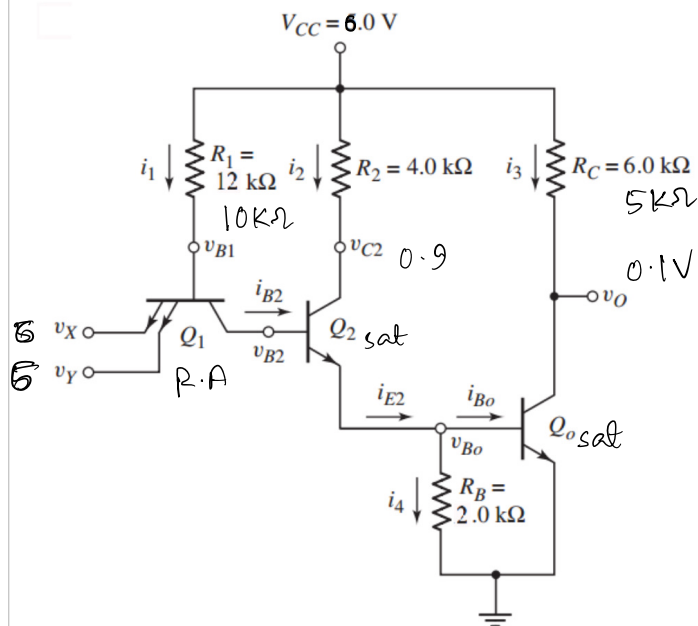
$$= 0.928 \times 40 = 0.037 \text{ A}$$

$$\text{So } I_C = \frac{V_{CC} - V_C}{10} \Rightarrow 0.037 = \frac{5 - V_C}{10}$$

$$\boxed{V_C = 4.629 \text{ V}}$$

$$\text{So } V_{CE} = 5 - (4.629 - 2.335) = 1.29 \text{ V} > 0.2$$

Question 02



a) $v_X = v_Y = 6V$ [Both input high]

$$i_1 = \frac{6 - 2.3V}{10k} = 0.37 \text{ mA} \quad (1)$$

$$i_{B2} = i_1 + i_{E2} + i_{E3} = 0.37 + 2 \times 0.1 \times 0.37 = 0.444 \text{ mA} \quad (2)$$

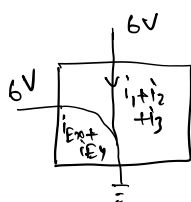
$$i_2 = \frac{6 - 0.9}{4k} = 1.275 \text{ mA} \quad (3)$$

$$i_{E2} = (1.275 + 0.444) = 1.72 \text{ mA}$$

$$i_{B3} = i_{E2} - i_4 = 1.32 \text{ mA} \quad (4)$$

$$(i_{E3}) \quad i_3 = \frac{6 - 0.1}{5k} = 1.18 \text{ mA} \quad (5)$$

b) Power dissipation for input = 6V [4 marks]



$$P = (6 - 0) \times (i_1 + i_2 + i_3 + i_{E2} + i_{E3}) = 6V \times 2.899 \text{ mA} = 17.394 \text{ mW} \quad (6)$$

c) Fanout Calculation, Any input of load = 0.1V

For any load input, $= 0.1V$

$$i_{EX} = i_L(\max) = i_1 = \frac{6 - 0.9}{10k} = 0.51 \text{ mA}$$

$$\frac{i_{CO}}{i_{BO}} < \beta_F \rightarrow i_{CO}(\text{no load}) + N \times i_L \text{ (is)} < \beta_F \times i_{BO}$$

$$\rightarrow 1.18 \text{ mA} + N \times 0.51 \text{ mA} < 30 \times 1.32 \text{ mA}$$

$$\rightarrow N < 75.33$$

$$\rightarrow \boxed{N = 75} \quad [5 \text{ marks}]$$

Fanout Calculation, both input of load = 0.1V

$$i_{EX} + i_{EY} = i_L(\max) = i_1/2 = 0.51/2 = 0.255 \text{ mA}$$

$$i_{CO} = i_3 + N \times i_L = i_{CO}(\text{no load}) + N i_L$$

$$\boxed{\beta_F = 30}$$

$$\frac{i_{CO}}{i_{BO}} < \beta_F \rightarrow i_{CO} < \beta_F \cdot i_{BO}$$

$$\rightarrow i_{CO} < i_{CO}(\max)$$

$$\Rightarrow i_{CO}(\text{no load}) + N i_L < i_{CO}(\max)$$

$$\Rightarrow 1.18 \text{ mA} + N \times 0.255 \text{ mA} < 30 \times 1.32 \text{ mA}$$

$$\Rightarrow N < 150.67 \rightarrow \boxed{N = 150} \quad [5 \text{ marks}]$$

①

$\Rightarrow$ For both input of load = 0.1V, load current reduced to half, which doubles fanout.

[1 mark]

Question No. 3

In **Figure 3** the circuit has the following parameters: $\beta_F = 30$, $V_{BE}(\text{Sat}/\text{Act}) = 0.7\text{V}$, $V_{CE}(\text{Sat}) = 0.1\text{V}$, $V_\gamma = 0.5\text{V}$, $V_D = 0.7\text{V}$, Zener breakdown voltage, $V_{ZN} = 5.1\text{V}$.

(a)	CO1	Briefly explain the advantages and disadvantages of HTL circuit over the modified DTL circuit.	[4]
(b)	CO2	When $V_A = 15\text{V}$ and $V_B = 15\text{V}$, determine the output voltage v_o and the operating modes of T1 , T2 . Repeat the process when $V_A = 0.1\text{V}$ and $V_B = 15\text{V}$ and determine V_o and the operating modes of T1 , T2 .	[6]
(c)	CO2	Considering the two input configurations mentioned in part (b), determine the noise margin of the HTL circuit.	[10]

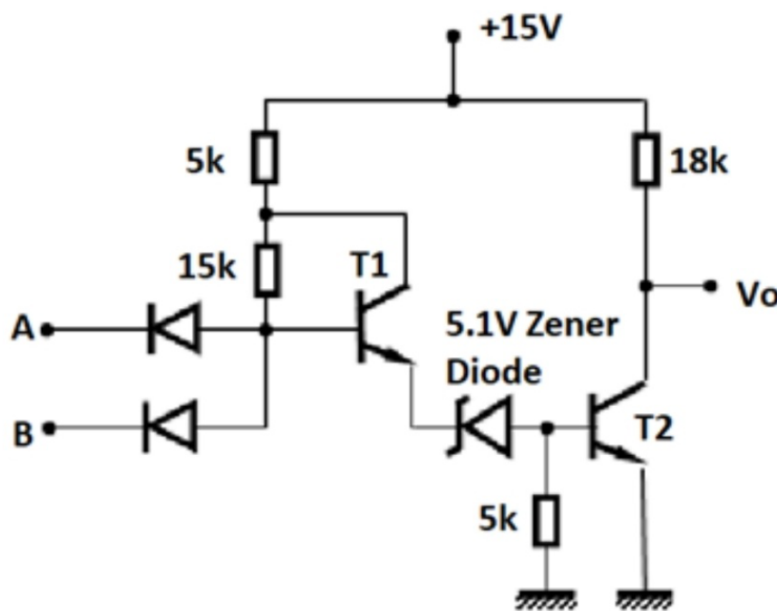


Figure 3

(a) Advantage: HTL circuits have very high noise margin and noise voltage levels. **(2 marks for advantage)**

Disadvantage: High operating voltages result in higher idle power draw and lower operating speeds. **(2 marks for dis)**

(b) **Case-1**

$$V_A = V_B = 15\text{V}$$

$$V_o = 0.1\text{V} \quad (1)$$

T₂ saturation (1)

T₁ forward active (1)

Case-2

$$V_A = 0.1\text{V}, V_B = 15\text{V}$$

$$V_o = 15\text{V} \quad (1)$$

T₁, T₂ cutoff (1+1)

(c)

Case-1

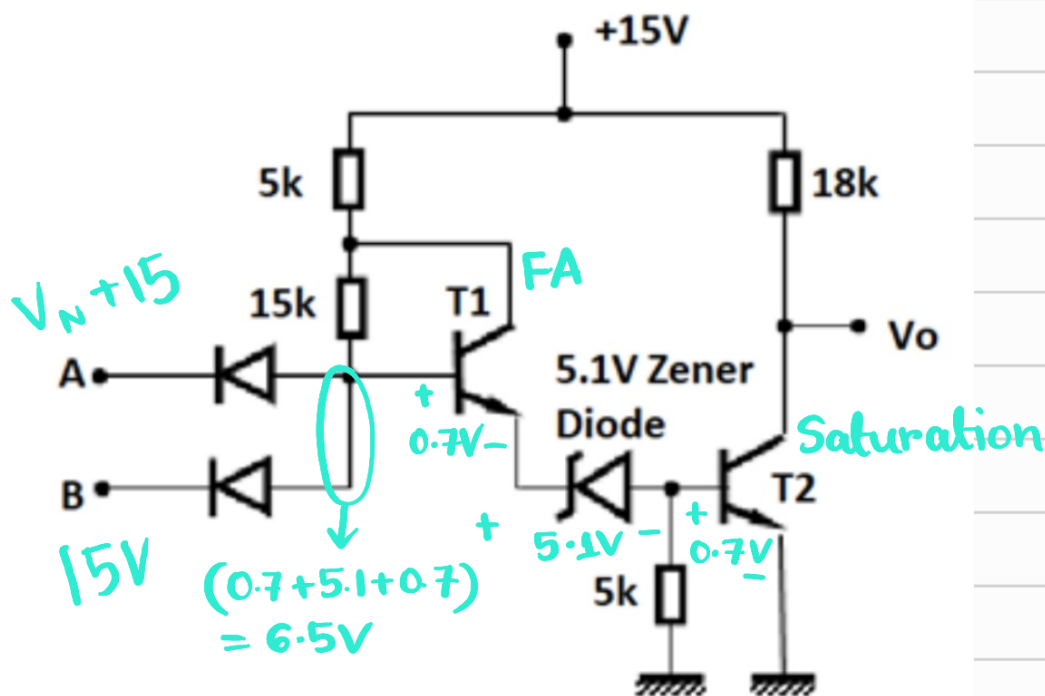


Figure 3

Original mode : D_A, D_B turned off

$$\begin{aligned} \text{node voltage} &= V_{BE}(\text{act}) + V_{ZN} + V_{BE}(\text{sat}) \\ &= 0.7 + 5.1 + 0.7 = 6.5V \end{aligned}$$

D_A will start to turn on if

$$\begin{aligned} V_N + 15 &\leq 6.5 - V_Y \\ V_N + 15 &\leq 6.5 - 0.6 \Rightarrow V_N = -9.1V \text{ [In case } V_Y = 0.6] \\ \Rightarrow V_N &\leq 6.5 - 0.5 - 15 = -9V \end{aligned}$$

$$\boxed{V_{NH} = 9V} \quad (4 \text{ marks})$$

Case-2

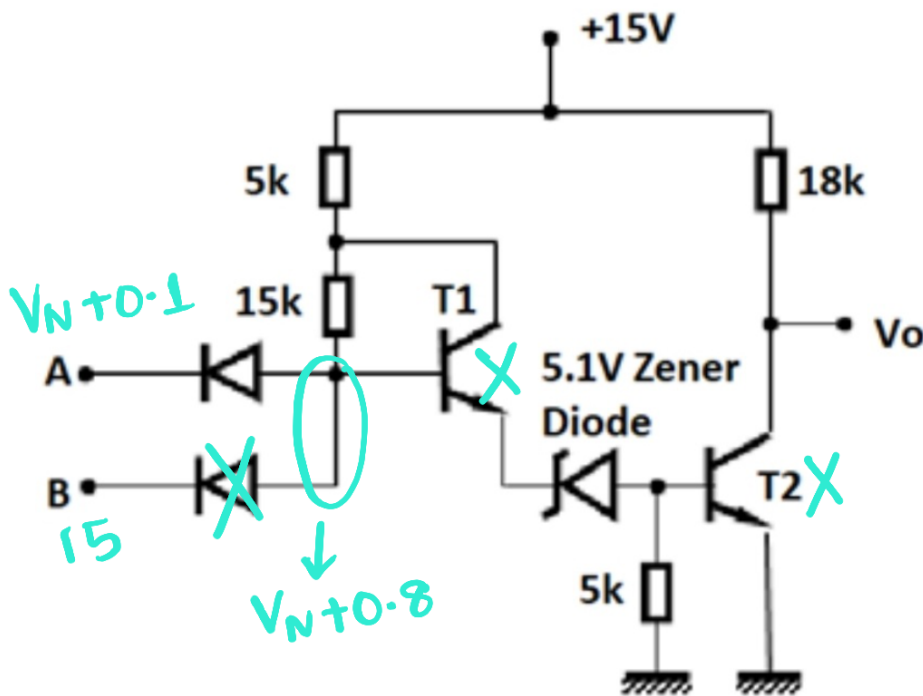


Figure 3

Original mode: D_A turned on, D_B, T_1, T_2 cutoff

$$\text{Node voltage} = V_N + 0.1 + V_D = V_N + 0.8V$$

$(T_1 - D_{ZN} - T_2)$ branch turn-on voltage

$$= (V_Y + V_{ZN} + V_Y) = 0.5 \times 2 + 5.1 = 6.1V$$

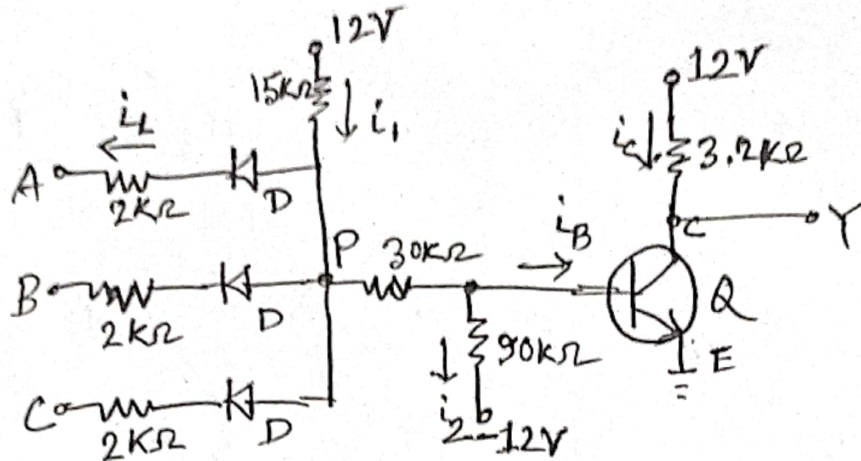
for successful operation $\Rightarrow V_N + 0.8 < 6.1V$

(5 marks)

$$\Rightarrow V_N < 5.3V$$

$$\therefore \boxed{V_{NL} = 5.3V} \quad \boxed{\text{Noise margin} = 5.3V} \quad (1 \text{ mark})$$

Q.4



→ ② ^{DTL} logical low input voltage = $0.2V [V_{CE(sat)}]$

assumption: All the diodes D's are ON/forward biased and transistor Q is in cutoff mode.

$$\therefore i_B = i_C = 0$$

Applying nodal analysis at node (P), (voltage at Base(B) may not be enough to turn on Q)

$$3 \times \frac{(V_P - 0.7) - 0.2}{2K} + \frac{V_P - 12}{15K} + \frac{V_P - (-12)}{120K} = 0$$

$$\Rightarrow \boxed{V_P = 1.3016V} \quad \leftarrow 5 \text{ marks}$$

$$\therefore i_1 = \frac{12 - 1.3016}{15K} = 0.7132 \text{ mA}$$

$$i_L = \frac{V_P - 0.9}{2K} = \boxed{0.2008 \text{ mA}} \quad \leftarrow 2 \text{ marks}$$

voltage at base (V_B) $\Rightarrow \frac{V_B - V_P}{30K} + \frac{V_B + 12}{90} = 0$

$$\rightarrow V_B = -2.024V$$

So, Q is in cutoff $\rightarrow$ assumption is correct

⑤ logical high input voltage = $V_{CC} = 12V$

assumption: Diodes D's are off/reverse biased.

Transistor Q is in saturation mode.

$$\therefore V_{BE} = 0.8V, V_{CE} = 0.2V$$

$$i_1 = \frac{12 - 0.8}{45K} = 0.248mA$$

$$i_2 = \frac{0.8 - (-12)}{90K} = 0.142mA$$

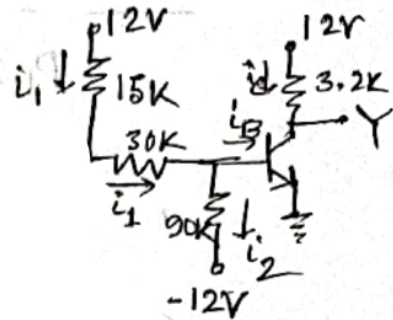
$$\therefore i_B = i_1 - i_2 = 0.106mA$$

$$i_C = \frac{12 - 0.2}{3.2K} = 3.6875mA$$

$$\therefore \beta_{Forced} = \frac{i_C}{i_B} = 34.7877 < \beta_{Forward} (=40)$$

$\therefore$ Transistor Q is in saturation mode and it is verified.

8 marks

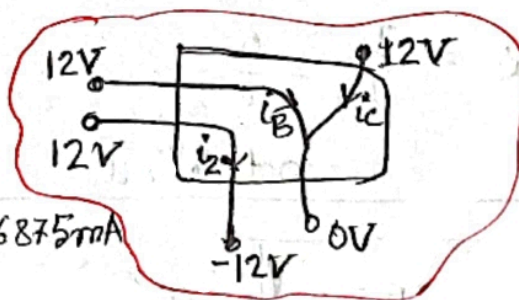


⑥

from (b),

$$i_2 = 0.142mA,$$

$$i_B = 0.106mA, i_C = 3.6875mA$$



power dissipation, $P = \sum (V \times I)$

$$= [12 - (-12)] \times i_2 + (12 - 0) \times (i_C + i_B)$$

$$= 24 \times 0.142 + 12 \times (0.106 + 3.6875)$$

$$= \boxed{48.93mW}$$

5 marks